Haocheng Zhu

Computational Cognition | *Multisensory Processing* | *Bayesian Modelling*
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RESEARCH PROFILE

Computational researcher developing probabilistic models of how intelligent systems perceive, attend, and reconstruct memory under uncertainty. Research spans recursive Bayesian filtering, nonparametric inference, and multimodal causal inference, with an emphasis on turning cognitive insights into generalizable computational methods.

EDUCATION

Ph.D. Candidate in Computational Cognition 2024-present
University of California, Los Angeles, USA Advisor: Ladan Shams, Ph.D.

M.A. in Computational Cognition 2024-2025
University of California, Los Angeles, USA

B.S. in Psychology 2020-2024
Soochow University, Suzhou, China

RESEARCH EXPERIENCE

Recursive Bayesian filtering framework for real-time multisensory causal inference

- Developed a recursive Bayesian filtering framework to fuse noisy multimodal sensory streams, jointly track a moving source, and infer whether sensory observations share a common cause.
- Achieved strong held-out predictive performance ($R^2 = 0.94$ vs. 0.87 for the baseline) and reproduced a temporal response signature unexplained by static inference or motor-latency controls
- Unified dynamic state estimation and causal inference within a single recursive framework, enabling an agent to continuously infer where a source is and which sensory signals belong together, which is a foundation for adaptive sensor fusion in embodied perception.

BCI Toolbox — open-source Bayesian modeling software

- Built and released a Python package with a graphical interface and programmatic API, translating Bayesian causal inference methods into reusable workflows for behavioral data modeling.
- Reached **66,000+ downloads**, bringing Bayesian causal inference methods to researchers through interactive and code-based workflows; published in *PLOS Computational Biology* (2024).

Attention-Guided Nonparametric Bayesian Inference for Open-World Multisensory Perception

- Pioneered an attention-guided nonparametric Bayesian framework that breaks beyond the fixed-source assumptions of classical causal inference, extending the Chinese Restaurant Process to reason over a dynamically expanding set of latent sensory sources.
- Unified attention and causal inference by making attention actively direct hypothesis selection, establishing a foundation for adaptive multimodal reasoning in complex environments with unknown numbers of competing sources.

Multidimensional Bayesian Modeling for Multimodal Sensor Fusion

- Developed a joint Bayesian inference framework that integrates evidence across multiple stimulus dimensions to infer shared causal structure, advancing beyond single-dimension models toward richer multimodal reasoning.
- Led model development and psychophysical validation, explaining 93% of behavioral variance with just 5 free parameters and outperforming forced-fusion and maximum-likelihood estimation baselines; published in *Current Biology* (2024) and *Psychonomic Bulletin & Review* (2026).

Computational Modeling of Memory Reconstruction and Suggestibility

- Brought causal inference model into memory research, introducing a computational account of reconstructive recall that reframes the misinformation effect as a consequence of rational inference under uncertainty.

- Validated the framework in a 308-participant study with UW–Madison, explaining 96% of group-level behavioral variance and outperforming four competing models on BIC; co-first-authored work accepted at *Communications Psychology*.

PEER-REVIEWED PUBLICATION

Parong, J.*, **Zhu, H.***, Kamal, K.*, Hang, L., Moreland, M., Shams, L., Seitz, A., & Green, C., S. (accepted). Memory Suggestibility: a 'Sin' or Optimal Causal Inference? ***Communications Psychology*** (*co-first)

Zhu, H., Beierholm, U., & Shams, L. (2026). Analyzing multisensory integration do's and don'ts. ***Neuroscience and Biobehavioral Reviews.*** *https://doi.org/10.1016/j.neubiorev.2026.107008*

Zhu, H., Zhang, Y., Beierholm, U., & Shams, L. (2026). Crossmodal interaction of flashes and beeps across time and number follows Bayesian causal inference. ***Psychonomic Bulletin & Review.*** https://doi.org/10.3758/s13423-026-02857-z

Wang, A., Zhang, J., Liu, G., **Zhu, H.**, Xing, Z., Yue, C., & Tang, X. (2025). Developmentally enhanced visual reliability reduces susceptibility to the sound-induced flash illusion: Evidence from Bayesian causal inference. ***Consciousness and Cognition***. https://doi.org/10.1016/j.concog.2025.103921

Zhu, H., Beierholm, U., & Shams, L. (2024). The overlooked role of unisensory precision in multisensory research. ***Current Biology***. https://doi.org/10.1016/j.cub.2024.01.057

Zhu, H., Beierholm, U., & Shams, L. (2024). BCI Toolbox: an open source package in python for Bayesian causal inference model. ***PLoS Computational Biology***. https://doi.org/10.1371/journal.pcbi.1011791

Zhu, H., Tang, X., Chen, T., Yang, J., Wang, A., & Zhang, M. (2023). Audiovisual illusion training improves multisensory temporal integration. ***Consciousness and Cognition***. https://doi.org/10.1016/j.concog.2023.103478

PUBLICATION SUBMITTED AND IN PREPARATION

Zhu, H., Beierholm, U., & Shams, L. (in prep). Selection before integration: attentional priority structures multisensory causal inference.

Wang, A., Qian, Q., **Zhu, H.,** Shen, L., & Chen, Q. (in prep). Neural dynamics in multisensory temporal integration: a f-MRI study.

CONFERENCE

Zhu, H. & Shams, L. (**symposium talk**) Bayesian causal inference in Multisensory Perception *Society for Neuroscience*, San Diego, CA, November, 2025

Zhu, H., Zhang, Y., Beierholm, U., & Shams, L. (talk) Crossmodal Interaction of Flashes and Beeps Across Time and Number Follows Bayesian Causal Inference. *International Multisensory Research Forum (IMRF)*, Durham, U.K., July, 2025

Zhu, H., Wang, A., Shams, L. (poster) Bayesian causal inference behind multisensory depth perception. *Chinese Computational Cognitive Neuroscience Conference (CCCN)*, HongKong, July, 2024

Zhu, H., Beierholm, U., & Shams, L. (poster) BCI Toolbox: an open source package in python for Bayesian causal inference model. *International Multisensory Research Forum (IMRF)*, Reno, U.S., August, 2024

Zhu, H., Zhang, Y, & Shams, L. (poster) Multidimensional Bayesian causal inference modulates multisensory perception. *Future of Neuroscience*, LA, U.S., November, 2024

Zhu, H., Wang, A., & Zhang, M. (poster) Audiovisual illusion training improves multisensory temporal integration. *6th China Vision Science Conference*, Wenzhou, P.R.C., August, 2023

ACADEMICS SERVICE

Workshop Co-Organizer, "Computational Models in Multisensory Perception," *International Multisensory Research Forum (IMRF)*, Durham, UK, July 2025

Reviewer for: *Nature Communications*, *Journal of Experimental Psychology: HPP; Perception; Acta Psychologica*; Frontiers in Human Neuroscience; Frontiers in Psychology; Science China: Life Sciences*

*Multiple times